

Online Appendix

Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

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OA.1 Algorithms

Algorithm OA.1 (amortized shape model). (i) Assemble the pooled panel $\{(s_{j,t}, \widehat{z}_{j,t})\}$ over names j and days t . (ii) Train H_ϕ by SGD on $\mathbb{E}_{\tau \sim \lambda} \rho_\tau(\widehat{z} - H_\phi(s, \tau))$ [network], or boost per- τ regressions on a grid [trees]; enforce monotonicity by rearrangement. (iii) For any asset, including one with no return history, using characteristics-only s , evaluate $H_\phi(s_t, \cdot)$; no per-asset estimation step exists. (iv) To *sample*, draw $\tau^{(m)} \sim U(0, 1)$ and return $z^{(m)} = H_\phi(s_t, \tau^{(m)})$.

Algorithm OA.2 (the misspecification meter, as deployed). (i) Fit equation (1) of the paper on a trailing window; store $\widehat{z}_{t-62}, \dots, \widehat{z}_t$. (ii) Compute mk_{63} , $|\text{sk}_{63}|$, J_5 by equation (12) of the paper. (iii) Convert each to its percentile against the trailing pooled panel (same asset class); the score is the maximum of the three percentiles, re-ranked into deciles. (iv) Act by the paper: top decile $\Rightarrow$ the parametric model’s average pinball loss runs $\sim 2\text{--}3\%$ higher within this score region (a conditional average, not a per-name point estimate), flag the asset for the nonparametric / residual-hybrid engine and for model-risk review; elsewhere $\Rightarrow$ no action (in the amortized setting the engine is run regardless; the meter serves surveillance, and day-level switching is unsupported, Section 5.4 of the paper). (v) Latency: detection lags episode onset by median 0 / mean 2–4 days; the edge is back-loaded within episodes, so the lag forfeits little.

The production loop. *Nightly:* (i) update each name’s GARCH filter state $\hat{\sigma}_{t+1}$ (recursion only; parameters refit annually); (ii) assemble the state vector s_t and evaluate the shared shape model $q_\phi(\tau | s_t)$, one model for the whole book, no per-name estimation; (iii) apply the frozen GPD tail below p_0 and, where the desk elects the coverage overlay, the conformal shift c_τ (static or adaptive); (iv) report $\hat{Q}_{t+1}(\tau) = \hat{\sigma}_{t+1} \tilde{q}(\tau | s_t)$ at the desk’s levels, with VaR and ES_{97.5} both read from the coherent min-envelope curve $Q^* = \min\{\text{body}, \text{EVT}\}$ (VaR at the α node, ES its numerical tail integral); (v) read the misspecification score equation (12) of the paper as the model-risk monitor: top decile means the parametric fallback’s conditional average loss in that score region runs $\sim 2\text{--}3\%$ higher. *Annually:* refit filters, retrain the shape model, re-estimate tail and shifts, all on trailing data. *New listing:* run the separate characteristics-only amortized forecaster until an own-history sample sufficient to initialize the per-name scale filter accrues; then attach this residual-hybrid pipeline. Total marginal cost per name per day: one filter update and one model evaluation.

OA.2 Supplementary figures

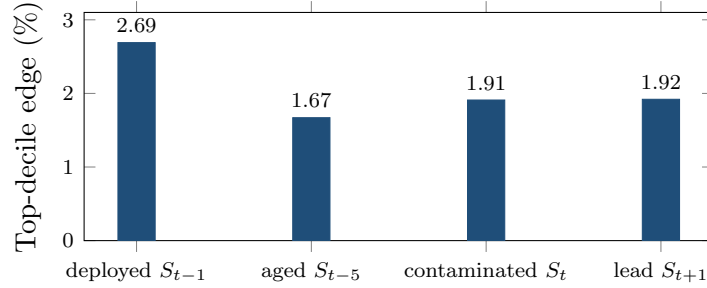


Figure OA.1: Timing diagnostics for the frontier. Top-decile edge under four alignments of the mk_{63} score with the daily loss (200 CRSP names; per-date DM 5.5–6.5 in all four). The deployed lag dominates, and the contaminated same-day alignment yields less, the opposite of what a look-ahead artifact would produce.

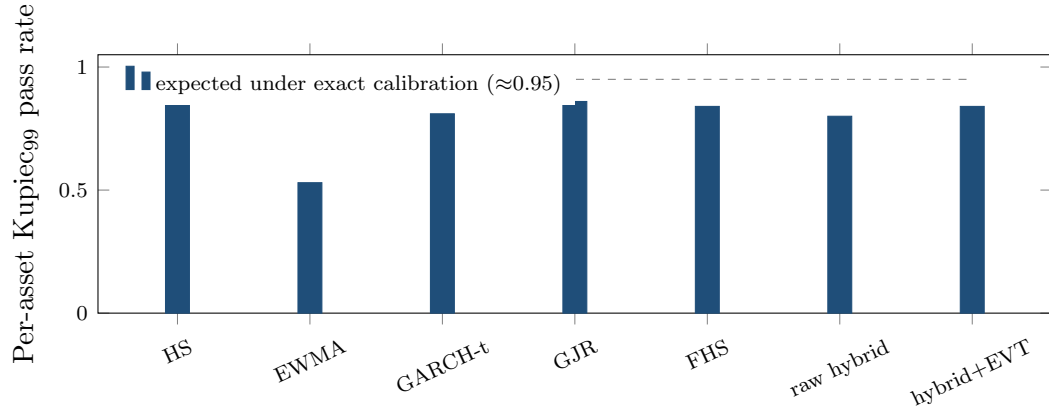


Figure OA.2: Exception tests restated per asset (140 CRSP names, 99% VaR): share of names whose individual Kupiec test is consistent with nominal. HS over-covers (conservative); EWMA fails; the EVT-tailed engine matches the parametric benchmarks while beating them on loss and the joint FZ0 score (Table 6 of the paper). Date-clustered tests in the table notes.

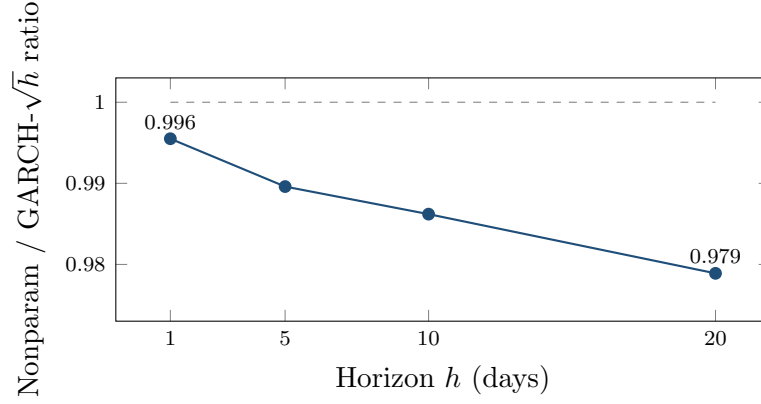


Figure OA.3: The edge grows with horizon. Direct-model/GARCH pinball ratio at $h = 1, 5, 10, 20$ days (150-name design panel; training labels whose h -day window crosses the split are purged): fat tails compound while $\sqrt{h}$ scaling compounds the shape error, from 0.45% at $h=1$ to 2.1% at $h=20$. The forecaster is the direct multi-day model of the main text’s ten-day subsection, *not* the residual-hybrid engine, whose architecture is one-day. An earlier version of this figure showed larger ratios from an unpurged run; the purged rerun supersedes it.

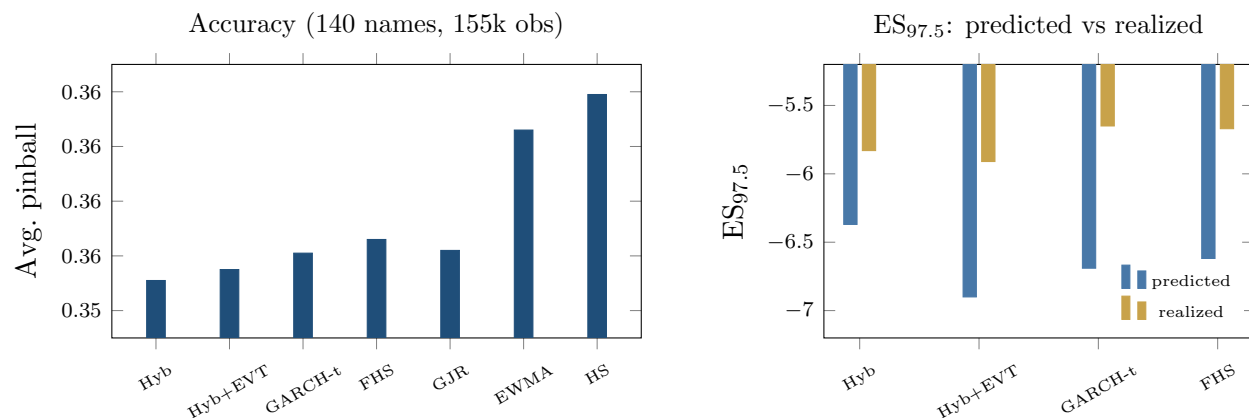


Figure OA.4: The FRTB battery. Left: average pinball loss by model (axis truncated to resolve the ordering; differences carry DM 5.2–8.8 vs the hybrid, all $p \approx 0$); CAViaR, not shown, statistically ties the hybrid. Right: ES_{97.5} predicted (exact tail integral) vs realized below each model’s own VaR, a model-dependent diagnostic, not a ranking (every fat-tailed entrant’s integral exceeds its own-tail mean); the joint FZ0 score of the paper is the ranking criterion.

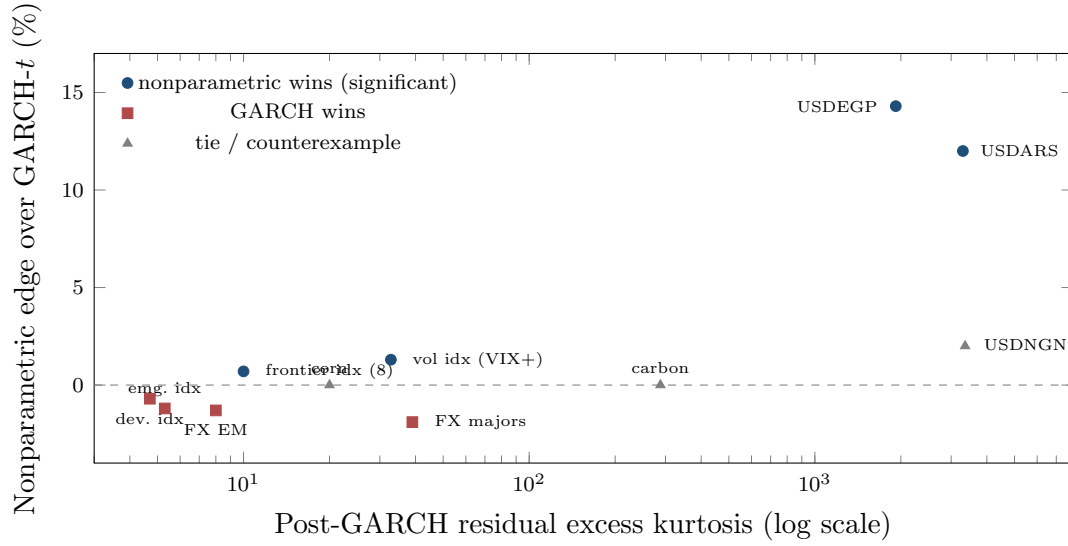


Figure OA.5: The frontier across universes: nonparametric edge vs post-GARCH residual kurtosis, tier means and single assets with study-reported kurtosis only (country universe corr 0.53; cross-asset corr 0.43). The hyperinflation-FX tier (kurtosis ~2000–3300) carries the only decisive standalone wins; high kurtosis is predictive but neither necessary nor sufficient: carbon (288) and corn (20) tie, and NGN’s win is not significant. The Korea-2026 control (kurtosis 4–13, GARCH wins all 23 assets) sits in the lower-left regime.

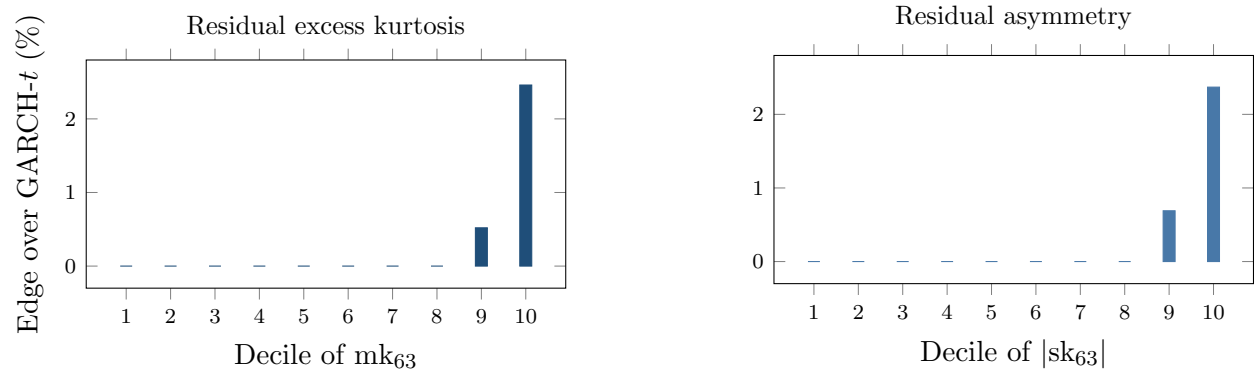


Figure OA.6: The misspecification frontier. Nonparametric pinball edge over GARCH- t by decile of the trailing 63-day residual excess kurtosis (left) and residual asymmetry (right), 200 CRSP names, 221.6k test asset-days. Deciles 1–8 are ≈ 0 (drawn at zero; exact bulk values in the result file); the edge jumps in the top decile (+2.46% kurtosis, +2.37% asymmetry; per-date DM 6.5 in the top kurtosis decile).

OA.3 Supplementary tables

Direct expected-shortfall backtests

Exception counts test the VaR level; they do not test ES. The main text adds two direct ES *calibration* backtests: the McNeil–Frey (2000) exceedance-residual test, whose statistic is the mean standardized loss beyond VaR, zero under a correct ES (McNeil and Frey, 2000); and the Acerbi–Székely (2014) statistic Z_2 , negative when ES understates the realized tail (Acerbi and Székely, 2014). These complement the jointly consistent FZ0 loss used for forecast comparison; Nolde and Ziegel (2017) and Bayer and Dimitriadis (2022) draw the distinction between backtesting the calibration of an ES forecast and comparing forecasts through a jointly consistent score. Table OA.5 reports the two calibration tests on the pooled panel (140 CRSP names, per-name GARCH- t fit on the first 60%, evaluated on the held-out 40%; 155,120 asset-day observations across 1,108 test dates), with inference clustered by calendar date. Gaussian innovations are decisively inadequate in this daily-data battery: the Gaussian filter is worst on every ES metric, its mean exceedance residual reaching -0.78 standard deviations at the 1% level ($Z_2 = -1.07$). Among the fat-tailed entrants no model dominates every diagnostic. The EVT-tailed engine carries the Z_2 closest to zero at both levels and is the only entrant not rejected by the 1% Kupiec exception test; at 2.5% it and FHS are the only models McNeil–Frey does not reject (engine mean exceedance residual -0.02 , $p = 0.49$), while GARCH- t and the Gaussian filter are. The limitation is the 1% tail: there every model’s mean exceedance residual is significantly negative, the engine included, and FHS’s is marginally closer to zero than the engine’s (-0.19 against -0.24), though both are rejected. The message is consistent with the exception tests and the joint score: fat-tailed specifications materially improve tail calibration, the engine’s tail is the best calibrated overall, and the 1% tail on large caps leaves room that no daily model here closes.

Realized measures and the scale–shape decomposition

The main-text battery reads only daily returns, the regime the estimator is built for. To place it against a method that reads intraday data, and to ask what the misspecification frontier is made of, we build five-minute realized measures from TAQ for thirty large-capitalization U.S. equities over 2014–2024 (realized variance is the sum of squared five-minute log-returns, cleaned by regular sale conditions, a daily price band, and a 20% return clip) and fit a Realized GARCH (Hansen et al., 2012): the log-linear conditional variance $\log h_t = \omega + \beta \log h_{t-1} + \gamma \log \text{RV}_{t-1}$ with a measurement equation $\log \text{RV}_t = \xi + \phi \log h_t + \tau(z_t) + u_t$ linking the realized measure to the latent variance, estimated jointly by Gaussian quasi-maximum likelihood. Table OA.6 reports the result on 1,108 common test dates. Two facts follow. The realized scale lowers FZ0 well below the daily-data accuracy core at both levels (DM 6.0 at 2.5%, 4.5 at 1%): intraday information prices volatility

Table OA.1: Summary of the paper’s results and scope conditions. Edges are percentage pinball reductions unless noted; DM statistics are per-date (conventions note of Section 5 of the paper), one-sided. Panel A: the deployed amortized residual-hybrid engine (worst case on the accuracy yardstick: no detectable loss; calm-cell 90% lower confidence bounds sit above -0.25%). A separate block reports the direct multi-day extension, which is not the engine. Panel B: boundary evidence from configurations not deployed, which motivates the design.

<i>Panel A: the engine</i>		
Result	Magnitude	Significance
vs GARCH- t / EWMA / HS, $h=1$ (avg)	+0.3 / +1.5 / +1.9%	DM 4.4–8.3
Top misspecification decile	+2.7%	DM 6.5
Hyperinflation FX (ARS, EGP)	+12–14%	DM 2.9–3.8
Frontier equity indices (4 of 8)	$\sim +1$ –3%	DM 2.1–3.5
2000–2013 frozen-spec holdout (untouched era)	top decile +0.9%	DM 2.2 (sort); recursive real-time rule +1.0%, DM 2.15; frozen-constant restatement in text
Point-in-time universe (Q1-2000 top 200, delistings kept)	top bucket +2.9%; overall +0.6%	DM 6.6 / 9.0, frozen real-time thresholds
Strict calendar splits; annual-refit walk-forward	top decile +2.5% under all three	DM 6.2 (2020 cut); 3.1 (overall, pre-GFC cut); 6.1 (walk-forward)
Familywise error over the 3×10 signal-decile grid	9 cells survive 5% FWER	Romano–Wolf step-down, incl. all three top deciles
Exception battery through the 2008 crisis	Kupiec 94% / Christof. 95% at 99%	date-clustered t 0.17 at 99%; at 97.5%, 0.35 (EVT tail) and -1.09 (static overlay, conservative); HS, EWMA fail
Joint (VaR, ES) FZ0 score, full panel	accuracy layer lowest at both levels	DM 4.8–4.9 vs GARCH- t , FHS at 1%, 5.1 at 2.5%; 6.2–9.6 vs GAS; static shift concedes 2.5% (DM -1.6), adaptive shift erases it (tie; top-decile cost remains)
Cold start / young listings	+6–10%	full-scale panel; no per-asset model applies
Calm-quintile edge	≈ 0 (point estimates +0.0–+0.2%)	bottom-decile 90% CIs lie above -0.25% ; no detectable loss
CAViaR (strongest academic rival)	tie (co-best in MCS)	DM 0.4
<i>Extension (a separate direct multi-day model, not the engine)</i>		
Ten-day horizon vs $\sqrt{h}$ -scaling	+1.4% (2014–24)	holds in 2020–22 stress; 2000–13 rerun: tie, DM 0.07 (era-dependent)
<i>Panel B: boundary evidence from configurations not deployed</i>		
Bare standalone neural net, single names	-4.5 to -6.2%	DM +12 to +18; motivates the hybrid design
Standalone nonparametric, calm single assets	GARCH as good or better	pooled DM -9.5 to -13.7 ; the shape edge needs misspecification
Korea 2026 crisis, 23 assets (falsification case)	GARCH wins all	DM -2.5 to -5 ; a price crash is not a shape break
mk ₆₃ -reweighted FHS (reweighting probe)	worse than plain FHS	DM -11 overall and top decile; sharp univariate conditioning collapses the sample
Baltic freight; IG credit ^a	+15%; +52%	DM 9.6; 30.6; accuracy only, calibration open

^a Credit and freight fail breach-independence (exceptions cluster); their entries are accuracy results only. The credit magnitude is from the FRTB-grade single-asset battery; the smaller 4% figure in the paper’s cross-universe table is from the separate lean hybrid-vs-GARCH sweep, different battery, both reported.

Table OA.2: Double sort: the edge needs both a shape break and an active regime. Cells are the nonparametric-over-GARCH- t pinball edge (%) in independent quintiles of the misspecification score (rows) and trailing volatility (columns), design-era test panel.

	rv ₆₃ Q1 (calm)	Q2	Q3	Q4	Q5 (active)
mk ₆₃ Q1 (low)	+0.24	+0.34	+0.10	+0.01	−0.14
Q2	+0.16	+0.20	+0.22	+0.09	−0.57
Q3	−0.02	+0.19	+0.25	+0.44	+0.04
Q4	+0.08	+0.15	+0.35	+0.23	+0.10
Q5 (high)	−0.10	+0.44	+0.68	+0.61	+2.33

Top-row date-clustered DM by volatility quintile: 0.7, 2.3, 1.2, 2.8, 6.4.
The bottom-left and top-right regions are the point: neither ingredient alone produces the concentration.

Table OA.3: Model selection by task and regime. Each row is established in the section cited.

Task / regime	Winner	Where
Point quantile accuracy, tabular state	GBM (trees)	Section 3.2 of the paper
Single asset, calm (~90% of asset-days)	GARCH- t	Section 5 of the paper
Single asset, top misspecification decile	Nonparametric (+2.7%, DM 6.5)	Section 5 of the paper
Amortized cross-name panel, any regime	Nonparametric always; no gate	Section 5.4 of the paper
Regulatory ES with a $\geq$ GARCH floor	Residual-hybrid + EVT + con-formal	Section 6 of the paper

more sharply than any daily filter. But the gain is scale, not shape. Holding the realized scale fixed and re-fitting the pooled state-conditioned residual-quantile learner, the EVT tail, and the misspecification score on the realized residuals moves FZ0 within noise, in magnitude below one Diebold–Mariano standard error overall and in the top realized-residual score decile (the top three deciles reach DM −1.4 at 1%, directional but not significant). On these large-capitalization names a realized scale absorbs most of what the score detects on daily-GARCH residuals, so on the intraday-covered segment the two are substitutes; the score’s distinctive value is in the daily-data regime and the cross-asset and day-one settings where a realized scale does not exist.

Table OA.4: Cross-universe multiplicity: Holm step-down over all 93 instrument-level comparisons in the three exploratory universes of the paper’s cross-universe table (24 FX pairs, 26 country indices, 43 cross-asset instruments). One-sided p from each instrument’s date-clustered DM statistic; familywise $\alpha = 5\%$. Ranks below twelve all have $p > 0.02$ and are omitted.

Rank	Instrument	Universe	DM	p (one-sided)	Holm cutoff	Verdict
1	PJM power ^a	cross-asset	283.1	$< 10^{-16}$	5.38×10^{-4}	artifact ^a
2	Baltic Dry	cross-asset	9.36	$< 10^{-16}$	5.43×10^{-4}	survives
3	VIX	cross-asset	5.71	5.7×10^{-9}	5.49×10^{-4}	survives
4	V2X	cross-asset	5.03	2.5×10^{-7}	5.56×10^{-4}	survives
5	VVIX	cross-asset	4.66	1.6×10^{-6}	5.62×10^{-4}	survives
6	USDEGP	FX	3.80	7.2×10^{-5}	5.68×10^{-4}	survives
7	Kenya NSE20	indices	3.52	2.2×10^{-4}	5.75×10^{-4}	survives
8	Wheat	cross-asset	3.51	2.2×10^{-4}	5.81×10^{-4}	survives
9	MOVE	cross-asset	3.22	6.4×10^{-4}	5.88×10^{-4}	fails
10	Sri Lanka CSEALL	indices	3.20	6.9×10^{-4}	5.95×10^{-4}	fails
11	Gasoline	cross-asset	3.20	6.9×10^{-4}	6.02×10^{-4}	fails
12	USDARS	FX	2.86	2.1×10^{-3}	6.10×10^{-4}	fails

The family is every instrument-level look in the three exploratory universes; the Korea-2026 falsification case (23 assets, GARCH- t wins throughout) is not part of the win family. Bonferroni at 0.05/93 gives the identical survivor set. The pooled crisis-FX comparison (DM 4.12, $p \approx 1.9 \times 10^{-5}$) and the kurtosis-edge correlations (0.53 across indices, 0.43 cross-asset) are pooled and gradient statistics outside this instrument-level family; they are the replication evidence the paper’s cross-universe claims rest on.

^a Retained in the family because the raw sweep contains it, but not a win: the paper’s cross-asset audit reclassifies electricity as a log-return artifact on near-zero and negative prices, refit on price differences, GARCH wins PJM (DM -2.73), so the survivor count the text quotes is seven.

Table OA.5: Direct expected-shortfall backtests. 140 CRSP names, pooled test panel. Breach is the realized exception rate (target α); Kupiec p is the unconditional-coverage test. Z_2 is the Acerbi–Székely (2014) statistic, negative when ES understates the tail. MF is the McNeil–Frey (2000) mean exceedance residual (standardized loss beyond VaR, zero under correct ES) with its bootstrap p .

Model	$\alpha = 2.5\%$				$\alpha = 1\%$			
	Breach	Kupiec p	Z_2	MF (p)	Breach	Kupiec p	Z_2	MF (p)
GARCH-normal	0.027	0.00	−0.36	−0.56 (0.00)	0.016	0.00	−1.07	−0.78 (0.00)
GARCH- t	0.028	0.00	−0.14	−0.07 (0.00)	0.011	0.03	−0.14	−0.29 (0.00)
GARCH-FHS	0.028	0.00	−0.14	−0.02 (0.39)	0.011	0.00	−0.18	−0.19 (0.00)
Engine (hybrid+EVT)	0.028	0.00	−0.12	−0.02 (0.49)	0.010	0.12	−0.12	−0.24 (0.00)

Inference for the McNeil–Frey and Z_2 p -values is clustered by calendar date over the 1,108 test dates, not pooled across the 155,120 asset-days, since the 140 names share within-day market shocks. Kupiec is the analytic unconditional-coverage test. Source: `es_backtests_results.json`.

Table OA.6: The scale–shape decomposition. Thirty large-cap U.S. equities, 2014–2024 (1,108 test dates), joint (VaR, ES) FZ0 loss (lower is better) at both FRTB levels. The scale is a proper Realized GARCH except in the first row (the daily-data accuracy core). DM is date-clustered (Newey–West, 5 lags) against the realized scale with an unconditional residual quantile; a positive DM is worse than that benchmark.

Model	FZ0 2.5%	DM	FZ0 1%	DM
Daily core (GARCH- t + EVT)	1.501	+6.0	1.774	+4.5
Realized scale + residual quantile	1.446	—	1.737	—
+ EVT tail	1.446	−0.2	1.740	+1.4
+ pooled shape learner	1.445	−0.2	1.739	+0.3
+ shape + EVT (realized-hybrid)	1.446	+0.1	1.736	−0.3

The daily core’s DM is against the realized benchmark (the intraday-scale gain). The pooled shape learner is the paper’s state-conditioned residual-quantile model, retrained on the realized-scale residuals. Returns are CRSP daily; the intersection is common across models. Source: `rgarch_bench_results.json`, `realized_hybrid_results.json`.

OA.4 The two-regime example

A synthetic boundary check: when the frontier does *not* appear

Before the real data, a simulation with known truth (run once, seed fixed) checks the formal results and, more usefully, delineates when the frontier edge should *fail* to appear. The data-generating process is deliberately favorable to the score: unit-variance Student- t residuals whose degrees of freedom switch between a calm regime ($\nu = 12$) and a fat regime ($\nu = 3$) under a persistent Markov chain; forecasters see the trailing 63-day kurtosis score and compete at $\tau \in \{0.01, 0.025\}$ (80,000 days; scale known, isolating shape). First, the Lemma 1 of the paper identity verifies numerically: the regret integral and the Monte-Carlo regret agree to five decimals (3.5×10^{-5} at $\tau = 0.01$). Second, the conformal shift moves realized coverage toward nominal on the held-out split, as Proposition 1 of the paper promises. Third, and most instructive, the score-binned nonparametric forecaster does *not* beat the fixed shape here: a single $t_{\hat{\nu}}$ fitted on the mixture ($\hat{\nu} = 6.8$) is nearly unbeatable at these levels, with total regret under 0.5% and no decile gradient. Two mechanisms explain it, and both teach. (i) By Lemma 1 of the paper regret is *quadratic* in the quantile wedge, and at $\tau \geq 0.01$ the wedge between the fitted compromise and either regime is small; static kurtosis *blending* is a second-order sin. (ii) The sample kurtosis of a t_{ν} law is a noisy statistic when tails are fat: its usual moment-based asymptotic variance requires a finite eighth moment ($\nu > 8$), and the population kurtosis itself does not exist for $\nu \leq 4$, so sixty-three observations cannot reliably rank regimes that differ only in a static ν . The jump and asymmetry signals, whose top deciles independently survive familywise control, corroborate the ordering, and a tail-index (Hill-type) variant of the score is an obvious robustness extension. For the empirical sections this means the real-data top-decile edge of Section 5 of the paper cannot be an artifact of stationary fat tails being averaged over (that configuration is the one the simulation shows a fixed shape handles) and is consistent with *local* shape breaks a single fitted ν cannot straddle: asymmetry, jump clustering, and residual misfit that co-move with the score. This is the same conclusion the Korea 2026 falsification case reaches from the opposite direction (price crashes are not residual misspecification), and it is why the score combines kurtosis with asymmetry and a jump indicator rather than kurtosis alone. (Simulation script `toy_example.py` ships with the code release.)

OA.5 Trading implications and scenario generation

Trading implications. The universes where the nonparametric tail wins (crisis FX, volatility indices, credit spreads, frontier equities) are the desks where tail-shape-aware sizing and hedging matter most. The *direction* of the trade matters as much as its location: a correct physical-measure tail forecast does not by itself make bought crash insurance profitable, because the crash

risk premium can survive conditioning on a correct forecast. The deployable side is the other one, *warehousing* (selling) the premium, with the model supplying tail control rather than alpha on the mean. The gating experiment provides no evidence that the score times individual crash days, so its role in a premium-harvesting book is continuous tail-risk measurement, not a market-timing signal.

Scenario generation. Because the shape stage carries an exact inverse-CDF sampler, calibrating intractable scenario generators (rough volatility, Hawkes, jump-diffusions) for CCAR/ICAAP-style stress design is a natural extension; it is not developed here.

OA.6 Proofs

The four formal results of Section 3.3 of the paper are stated with proof sketches in the body; the complete arguments, with all regularity hypotheses made explicit, are collected here. Throughout, $\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\})$ and $q_F = F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}$ is the lower generalized inverse.

Proof of Lemma 1 of the paper (pinball regret identity)

Assume $\mathbb{E}|Z| < \infty$, so

$$S(q) = (1 - \tau) \int_{(-\infty, q]} (q - z) dF(z) + \tau \int_{(q, \infty)} (z - q) dF(z)$$

is finite for every q , and S is convex (an expectation of the convex maps $q \mapsto \rho_\tau(z - q)$). For fixed z , $q \mapsto \rho_\tau(z - q)$ is Lipschitz with constant $\max(\tau, 1 - \tau)$, so the difference quotients $[\rho_\tau(Z - q') - \rho_\tau(Z - q)]/(q' - q)$ are bounded by the integrable constant $\max(\tau, 1 - \tau)$; dominated convergence justifies differentiating under the expectation wherever the one-sided limits exist. Off the atoms of F ,

$$\frac{d}{dq} \rho_\tau(z - q) = \mathbb{I}\{z < q\} - \tau, \quad \implies \quad S'(q) = \mathbb{E} \mathbb{I}\{Z < q\} - \tau = F(q^-) - \tau,$$

which equals $F(q) - \tau$ for all but countably many q . Since S is convex, hence absolutely continuous, the fundamental theorem of calculus gives

$$S(q) - S(q_F) = \int_{q_F}^q (F(u) - \tau) du,$$

the countable atom set being Lebesgue-null; *no continuity hypothesis on F is needed for this identity*. For nonnegativity, note $q_F = \inf\{x : F(x) \geq \tau\}$ implies $F(u) \geq \tau$ for $u \geq q_F$ and $F(u) < \tau$ for $u < q_F$. If $q > q_F$ the integrand is ≥ 0 on $[q_F, q]$, so the integral is ≥ 0 ; if $q < q_F$,

$$S(q) - S(q_F) = \int_{q_F}^q (F(u) - \tau) du = \int_q^{q_F} (\tau - F(u)) du \geq 0,$$

the integrand again ≥ 0 on $[q, q_F]$. Equality holds iff $F(u) = \tau$ for Lebesgue-a.e. u between q_F and q . Finally, if F is absolutely continuous with density $0 < m \leq f \leq M$ on the interval joining q_F and q , then F is continuous there, so $F(q_F) = \tau$ and $F(u) - \tau = \int_{q_F}^u f$; hence $m|u - q_F| \leq |F(u) - \tau| \leq M|u - q_F|$, and integrating over the interval yields $\frac{m}{2}(q - q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2}(q - q_F)^2$. $\square$

Proof of Proposition 1 of the paper (split-conformal validity)

Let $k = \lceil (n+1)\tau \rceil$ and assume $1 \leq k \leq n$ so that the k -th order statistic $e_{(k)} = c_\tau$ of the n calibration errors $\{e_1, \dots, e_n\}$ exists; when $k = n+1$ (i.e. τ within $1/(n+1)$ of 1) set $c_\tau = +\infty$, giving coverage 1. Suppose the $n+1$ errors $e_1, \dots, e_{n+1}$ (calibration plus the test error $e_{n+1} = z_{n+1} - \widehat{q}(\tau \mid s_{n+1})$)

are exchangeable and almost surely distinct. Then the rank R of e_{n+1} among the pooled $n+1$ values is uniformly distributed on $\{1, \dots, n+1\}$. By construction $\tilde{q}(\tau \mid s_{n+1}) = \hat{q}(\tau \mid s_{n+1}) + c_\tau$, so

$$\{z_{n+1} \leq \tilde{q}(\tau \mid s_{n+1})\} = \{e_{n+1} \leq c_\tau\} = \{e_{n+1} \leq e_{(k)}\} = \{R \leq k\},$$

the middle equality using a.s. distinctness (with ties the middle equality fails; e.g. if all e_i are equal the event has probability 1, which can exceed the stated upper bound). Therefore

$$\Pr(z_{n+1} \leq \tilde{q}(\tau \mid s_{n+1})) = \Pr(R \leq k) = \frac{k}{n+1} = \frac{\lceil (n+1)\tau \rceil}{n+1}.$$

Since $(n+1)\tau \leq \lceil (n+1)\tau \rceil < (n+1)\tau + 1$, dividing by $n+1$ gives $\tau \leq k/(n+1) < \tau + \frac{1}{n+1}$. The probability is over the joint law of calibration and test points; the guarantee is marginal, not conditional on s_{n+1} , because a single scalar shift c_τ pooled across states cannot deliver state-conditional coverage. $\square$

Proof of Proposition 2 of the paper (tail wedge)

Fix $\nu > 2$, so the t_ν law has finite variance and its unit-variance version $Z = T_\nu \sqrt{(\nu-2)/\nu}$ is well defined. The t_ν density is $\sim C_\nu |x|^{-(\nu+1)}$ as $|x| \rightarrow \infty$, so the lower tail is regularly varying with index $-\nu$: $\Pr(T_\nu < -x) = \tilde{c}_\nu x^{-\nu}(1 + o(1))$. Unit-variance scaling multiplies the constant, $\Pr(Z < -x) = c_\nu x^{-\nu}(1 + o(1))$ with $c_\nu = \tilde{c}_\nu((\nu-2)/\nu)^{\nu/2}$, and leaves the index ν unchanged. Writing $U(\tau) = |Q_\nu(\tau)|$ and inverting the regularly varying tail (Embrechts et al., 1997) gives $\tau = c_\nu U(\tau)^{-\nu}(1 + o(1))$, hence $U(\tau) = (c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$, i.e. $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$. For $\nu_1 < \nu_2$,

$$\frac{|Q_{\nu_1}(\tau)|}{|Q_{\nu_2}(\tau)|} = \frac{(c_{\nu_1}/\tau)^{1/\nu_1}}{(c_{\nu_2}/\tau)^{1/\nu_2}} (1 + o(1)) = \text{const} \cdot \tau^{-(1/\nu_1 - 1/\nu_2)} (1 + o(1)) \rightarrow \infty$$

because $1/\nu_1 > 1/\nu_2$. Thus for all sufficiently small τ both quantiles are negative with $|Q_{\nu_1}(\tau)| > |Q_{\nu_2}(\tau)|$, i.e. $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \rightarrow \infty$. The ordering therefore holds for every $\tau < \delta$ for some $\delta > 0$; putting $\tau^* = \min(\delta, \frac{1}{2})$ gives $\tau^* \in (0, \frac{1}{2})$ as asserted. (This establishes *existence* of τ^* ; the actual crossover τ_c where the ordering flips is not claimed here and is located numerically in the crossover remark of the paper.) Under local truth t_{ν_1} , applying the corresponding equation of the paper with $q = Q_{\nu_2}(\tau)$ and $q_F = Q_{\nu_1}(\tau)$ bounds the regret below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$, with m the minimum of the t_{ν_1} density on the wedge interval. $\square$

On the score-ordering: why it is an empirical regularity, not a theorem

We state the score-ordering as an empirical regularity rather than a theorem, and record here why no clean monotonicity theorem holds at the deployable levels. The tail-wedge result (Proposition 2

of the paper) orders a given pair of Student- t tail indices only below a pair-specific crossover level $\tau^*(\nu_1, \nu_2)$; it does not deliver monotonicity of the wedge in the tail index at a fixed operational τ . More fundamentally, the innovations are unit-variance by construction, and for the unit-variance t_ν the level- τ quantile is $Q_\nu(\tau) = \sqrt{(\nu-2)/\nu} t_\nu^{-1}(\tau)$, which is *not* monotone in ν : as $\nu \downarrow 2$ the normalization $\sqrt{(\nu-2)/\nu} \rightarrow 0$ pulls the fixed- τ quantile back toward zero, so an arbitrarily heavy-tailed unit-variance t eventually has a fixed- τ quantile arbitrarily close to the center rather than far into the tail. The non-monotonicity is already active at the regulatory levels: at $\tau = 0.025$, $Q_5(0.025) \approx -1.99$ is *less* extreme than $Q_8(0.025) \approx -2.00$, so a heavier-tailed standardized t carries the less extreme 2.5% quantile. A clean ordering theorem would therefore have to restrict the tail index to an explicit interval, isolate a range of τ on which $\partial Q_\nu(\tau)/\partial \nu$ has one sign, and control the density factor in the regret expansion; we do not pursue that narrow statement. The exact content we retain is the regret identity (Lemma 1 of the paper) and the tail-wedge ordering (Proposition 2 of the paper), and the frontier itself is documented empirically in the paper. $\square$

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Proof of Lemma 2 of the paper (generative sampler)

Let $\tilde{q}(\cdot \mid s)$ be nondecreasing and left-continuous on $(0, 1)$ and $\tau \sim U(0, 1)$, and set $Z^* = \tilde{q}(\tau \mid s)$ and $G(x) = \sup\{t \in (0, 1) : \tilde{q}(t \mid s) \leq x\}$ (with $\sup \emptyset = 0$). Monotonicity makes $\{t : \tilde{q}(t \mid s) \leq x\}$ an interval with left endpoint 0, and left-continuity ensures the supremum is attained when the set is nonempty, so the set equals $(0, G(x)]$; hence $\{Z^* \leq x\} = \{\tau \leq G(x)\}$ and $\Pr(Z^* \leq x) = G(x)$, i.e.

Z^* has CDF G . Because G so defined is the upper generalized inverse of $\tilde{q}$, and a left-continuous nondecreasing $\tilde{q}$ is in turn the lower generalized inverse of G , the quantile function of Z^* is exactly $\tilde{q}(\cdot \mid s)$. When $\tilde{q}$ is an estimated curve, the argument applies verbatim to the estimate: the sampler is exact for the *fitted* conditional law, which is the claim. $\square$